

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 4: Heat

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	4 — Heat
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps baited here: equal-amount mixing must land midway while unequal amounts pull the mix toward the larger share; 'hotter = more heat' and 'feels colder = is colder' reflexes; division-size misreads where the small marks are NOT tenths; conductor/insulator role-swaps; emitter-vs-absorber colour confusion; breeze-direction reversals tied to which surface heats or cools faster. Do not write on this paper — mark answers on your answer sheet.

1. 200 g of water at 20 °C is mixed with 200 g of water at 80 °C in an insulated jar (no heat escapes). Using only the idea that equal amounts share the gap equally, the final temperature is closest to:

- (A) 80 °C, because the hotter water decides the final temperature
- (B) 100 °C, because the two temperatures add together (20 + 80)
- (C) 50 °C, midway between 20 and 80, because equal amounts of the same liquid meet halfway
- (D) 20 °C, because heat always settles at the lower of the two values

2. 300 g of water at 80 °C is mixed with only 100 g of water at 20 °C in an insulated jar. Without any formula, the soundest estimate of the final temperature is:

- (A) Exactly 50 °C, the midpoint of 20 and 80, regardless of the amounts
- (B) Below 50 °C, nearer 20 °C, because cold water always wins
- (C) Exactly 60 °C, because you simply average the two temperatures $(20 + 80) \div 2 = 50$
- (D) Above 50 °C — much nearer 80 °C — because the larger 300 g of hot water dominates the mix

3. A tiny drop of boiling water at 100 °C falls into a large bucket of water at 25 °C. The student must predict the bucket's final temperature. The best reasoning is:

- (A) It stays almost exactly 25 °C, because the bucket holds far more water and a single drop carries very little heat
- (B) It rises to about 62.5 °C, the midpoint of 25 and 100
- (C) It rises sharply because the drop is the hottest thing present
- (D) It drops below 25 °C because adding water always cools a container

4. Block X has twice as much metal as block Y, and both are the same metal. X is at 40 °C and Y is at 40 °C. They are pressed together. What happens to their temperatures?

- (A) X cools and Y warms, because the bigger block always gives heat to the smaller one
- (B) Nothing changes — both stay at 40 °C, because there is no temperature difference to drive any heat flow
- (C) Y warms toward X because more metal means a higher temperature
- (D) Heat flows from X to Y because larger objects hold more heat and must release it

5. A cup of tea is left in a room at 25 °C and slowly cools. A student says, 'The room is warming up the tea a little while the tea cools.' What is the correct description of the energy flow?

- (A) Heat flows both ways at once, so neither the tea nor the room changes
- (B) Cold flows from the room into the tea, which is what cools the tea
- (C) Heat flows from the hot tea into the cooler room air; the tea loses heat and the room gains a tiny amount, until the tea reaches room temperature
- (D) The room heats the tea because larger spaces always pass heat to smaller objects

6. Equal amounts of water are warmed: sample P from 20 °C to 30 °C, and sample Q from 70 °C to 80 °C. Each gained 10 °C. Which statement is the most accurate?

- (A) Q took in more heat, because it ended at the higher temperature
- (B) P took in more heat, because starting cold means it needs more heat to warm
- (C) Each sample took in roughly the same amount of heat, since equal amounts of the same liquid rose by the same number of degrees
- (D) No heat entered either sample, because heat and temperature are unrelated

7. A clinical thermometer reads only about 35 °C to 42 °C, while a laboratory thermometer reads about –10 °C to 110 °C. A technician must (i) record the freezing point of a salt solution near –5 °C and later (ii) note a feverish child's temperature to a tenth of a degree and carry the reading to another room. The correct pairing is:

- (A) Use the clinical thermometer for both tasks, since it reads in Celsius
- (B) Use the laboratory thermometer for both tasks, since its wide range covers everything

- (C) Use the laboratory thermometer for the -5°C salt solution and the clinical thermometer for the carried-away fever reading
- (D) Use the clinical thermometer for the -5°C solution and the laboratory thermometer for the fever

8. A nurse records 39.4°C from a feverish patient but forgets to shake the thermometer down. Half an hour later she places the same thermometer (still unshaken) under a different patient whose true temperature is 38.6°C . What does she read, and why?

- (A) About 39.4°C , because the kink holds the thread at the earlier higher value and a 38.6°C body is not warm enough to push it higher
- (B) About 38.6°C , because the thermometer resets itself to each new patient
- (C) About 41.0°C , because the two readings add together
- (D) About 38.6°C , because the thread always falls back to the room temperature between uses

9. Why would a laboratory thermometer be a poor choice for confirming a fever of 39.2°C , even though 39.2°C lies well inside its -10°C to 110°C range?

- (A) A fever of 39.2°C is above its range, so it cannot reach that temperature
- (B) Its scale is spread over a huge range, so its smallest divisions are coarse (about 1°C) and it cannot resolve a tenth of a degree the way a clinical thermometer can
- (C) It has no kink, so it cannot measure body temperature at all
- (D) Its liquid freezes near body temperature, giving a wrong reading

10. A laboratory thermometer is placed in boiling water at 100°C . A student tips it almost horizontal to read it more easily, then pulls it out into the cooler air. Compared with a clinical thermometer treated the same way, what happens to the displayed reading?

- (A) Both readings stay locked, because every thermometer traps its thread once it stops rising
- (B) Only the laboratory thermometer holds its reading, because it has no kink to interfere
- (C) The laboratory thermometer's thread falls as the liquid cools in the air, so its reading changes; a clinical thermometer's kink would hold its reading instead
- (D) Both readings keep climbing once removed, because the glass stays hot

11. Two thermometers are both marked 0°C to 100°C over the same length of glass. Thermometer A has 100 small divisions across that span; thermometer B has 50. Which can be read more finely, and what is each smallest division worth?

- (A) Thermometer B reads more finely, because fewer divisions each cover a smaller temperature
- (B) Both read equally finely, because they cover the same $0-100^{\circ}\text{C}$ range
- (C) Thermometer A's divisions are 0.5°C and B's are 1°C
- (D) Thermometer A reads more finely; each of its divisions is 1°C , while each of B's is 2°C

12. A clinical thermometer is spread from 35°C to 42°C . Between each whole degree there are 10 equal small divisions. A patient's thread stops on the 7th small division above the 38 mark. The reading is:

- (A) 38.07°C
- (B) 38.7°C
- (C) 45°C
- (D) 38.35°C

13. On a laboratory thermometer the numbered marks are 20 degrees apart, with 10 equal small divisions between consecutive numbers. The thread rests on the 6th small division above the 40 mark. The temperature is:

- (A) 46°C
- (B) 52°C
- (C) 40.6°C
- (D) 60°C

14. A laboratory thermometer is numbered every 5 degrees with 5 equal divisions between numbers. Yesterday the thread sat on the 4th division above the 10 mark; today it sits on the 2nd division above the 15 mark. By how much did the temperature change?

- (A) It rose by 2°C , from 14°C to 16°C
- (B) It rose by 3°C , from 14°C to 17°C
- (C) It fell by 2°C , because today's division count (2) is smaller than yesterday's (4)
- (D) It rose by 5°C , the gap between the 10 and 15 marks

15. A laboratory thermometer numbered every 10 degrees with 10 divisions between numbers shows the thread 7 divisions below the 0 mark (in the negative region). What temperature does it read?

- (A) 7°C
- (B) -70°C
- (C) -17°C
- (D) -7°C

16. Two students read the same laboratory thermometer where the numbered marks are 2 degrees apart with 4 equal divisions between them. The thread is on the 3rd division above the 24 mark. One writes 24.3°C , the other writes 25.5°C . Who is right and why?

- (A) 24.3°C is right, because 3 divisions above a mark always means three-tenths
- (B) Both are right, since 24.3 and 25.5 are just two ways of writing the same value
- (C) 25.5°C is right — each division is 0.5°C , so 3 divisions above 24 is $24 + 1.5 = 25.5^{\circ}\text{C}$
- (D) Neither is right; 3 divisions above 24 is 27°C

17. A laboratory thermometer's thread climbs steadily during heating. At 9:00 it read the 2nd division above 30; at 9:01 the 5th division above 30; at 9:02 the 8th division above 30. Each division is 1°C . If the same steady rate continues, what will it read at 9:03?

- (A) About 39°C , because the thread cannot rise past the 9th division above 30
- (B) About 48°C , because the temperature doubles each minute
- (C) About 41°C , since it rises 3°C each minute (32, 35, 38, then 41 at 9:03 — i.e. the 1st division above 40)
- (D) Exactly 40°C , because the thread always stops at a numbered mark

18. A long metal rod has its left end in a flame. Identical wax beads are stuck at equal spacing all along it. The beads fall off one by one, starting from the flame end and moving toward the cool end, each a little later than the one before. The clearest conclusion about how heat moves AND why the timing is gradual is:

- (A) Heat radiates equally to every bead at once, so they should all fall together — the gradual order is just chance
- (B) Hot metal physically rises and carries heat bead to bead, which is convection inside the solid rod
- (C) Each bead generates its own heat after it warms, so the order shows the beads igniting in turn
- (D) Heat conducts particle-to-particle along the rod, arriving later at points farther from the flame, so distant wax melts later

19. Three identical rods — copper, glass, and wood — each have a wax bead on the far end and are heated equally at the other end. The copper bead drops in 30 seconds, the glass bead in about 5 minutes, and the wood bead barely loosens after 15 minutes. Ranking the three from best to worst heat conductor, the correct order is:

- (A) Wood, then glass, then copper, because lighter materials carry heat fastest
- (B) Glass, then copper, then wood, because glass is the best conductor of the three
- (C) Copper, then glass, then wood — the rod whose far bead melts soonest conducts heat fastest
- (D) All three conduct equally well, since the heat source was identical for each

20. A copper rod and a wooden rod of identical shape are both heated at one end. To make the far end of the wooden rod reach the same temperature the copper rod's far end reached in 1 minute, the wooden rod must be heated:

- (A) For exactly 1 minute, since both rods are the same shape and size
- (B) For less than 1 minute, because poor conductors heat up faster
- (C) For much longer than 1 minute, because wood is a poor conductor and passes heat to its far end far more slowly
- (D) Never, because wood cannot conduct any heat at all

21. On a cold morning a metal doorknob and a wooden door panel are both at the same room temperature of 10 °C. You touch each with a 30 °C hand. From which does your hand lose heat faster, and which feels colder?

- (A) The wood feels colder, because wood is at a lower temperature than the metal
- (B) Your hand loses heat faster to the metal knob, so the metal feels colder, even though both are at the same 10 °C
- (C) Both feel equally cold, since both are at the same 10 °C
- (D) The metal feels warmer, because metal stores more heat and gives some back to the hand

22. Why does conduction transfer heat quickly through a copper block but extremely slowly through the same-sized block of still air, in terms of particles?

- (A) Air contains no particles, so there is nothing in it to carry heat by conduction
- (B) Copper is colder than air, and colder materials always conduct better
- (C) Air's particles move too slowly to carry any energy, so conduction stops
- (D) Copper's particles are packed closely and pass on vibrations to neighbours readily, while air's particles are far apart and rarely collide, so they hand on energy very slowly

23. A cook wants the food in a pan to heat quickly from the burner, but wants the handle to stay cool to hold. Which combination of materials achieves BOTH goals?

- (A) A metal pan body (good conductor, passes heat into the food) with a wooden or plastic handle (poor conductor, blocks heat from reaching the grip)
- (B) A wooden pan body and a metal handle, so heat is kept out of the food and led safely along the handle
- (C) A metal body and a metal handle, since one material is simplest and metal handles stay coolest
- (D) A plastic body and a plastic handle, so the whole pan resists heat and never gets dangerously hot

24. Water in a kettle is heated only at the centre of its base. Just above the flame, the water:

- (A) Becomes denser and sinks, while cool water rises over it, because hot water is always the heaviest
- (B) Stays still and simply conducts its heat straight up through the layers above without moving
- (C) Turns instantly to steam at the bottom and bubbles straight up without any circulation
- (D) Expands, becomes less dense, and rises, while cooler denser water from the sides sinks to take its place, setting up a circulating current

25. A beaker of water is heated at one corner of its base, not the centre. Coloured dye dropped at that corner traces a loop: up over the heated corner, across the top, and down the far side. What does this reveal about where convection currents form?

- (A) The current always rises in the exact centre of the beaker, no matter where the heat is applied
- (B) There is no current; the dye only spreads by conduction through the still water
- (C) The dye sinks first at the heated corner because heating makes water heavier there
- (D) The current rises directly over wherever the water is heated and sinks where it is cooler, so an off-centre heat source still drives a loop rooted at that corner

26. A room is heated by a heater placed near the floor. Smoke from an incense stick beside the heater rises, crosses the ceiling, cools at the far wall, and sinks down it. If the same

heater were instead hung near the ceiling, what would most likely happen to the room's heating?

- (A) The room would heat faster, because heat from up high falls down to fill the room
- (B) Nothing would change, because air carries heat equally well from any height by conduction
- (C) The floor would become the hottest part, because warm air always sinks to the floor
- (D) The warm air from a ceiling-level heater would stay near the ceiling, so the lower part of the room would be poorly heated — which is why heaters are placed low

27. A student claims, 'If I heat a solid iron bar in the middle, the warm part will rise and cold parts will sink, just like water in a kettle, setting up a convection current inside the bar.' Why is this impossible?

- (A) It is possible, but only if the iron bar is first painted black to absorb the heat
- (B) It is possible, because any material can convect once it is hot enough
- (C) Convection needs a fluid whose heated portions can physically move and circulate; in a solid bar the particles are fixed in place and cannot flow, so heat moves only by conduction
- (D) It is impossible because a solid bar cannot be heated at all in the middle

28. A tall glass cylinder is set up with very hot coloured water gently floated on top of cold clear water below, arranged so they do not mix. Almost no convection current develops. The best explanation is:

- (A) The lighter warm water is already on top and the denser cold water below — the stable arrangement — so there is no buoyancy difference to drive circulation
- (B) Warm water is denser, so it should be at the bottom; having it on top stops all heat transfer
- (C) Convection cannot occur in any vertical cylinder, only in wide flat pans
- (D) The cold water on the bottom blocks the warm water from ever cooling

29. On a still, sunny day, smoke from a small fire over a dark patch of bare ground climbs almost vertically, while smoke over a nearby shaded grassy patch barely rises. The difference is best explained by:

- (A) Smoke is naturally lighter over bare ground than over grass, regardless of any heating
- (B) The dark bare ground gets hotter, warming the air above it more, so that air rises faster as a stronger convection current, lifting the smoke higher
- (C) Conduction through the ground physically pushes the smoke upward over the dark patch
- (D) Radiation from the bare ground grabs the smoke particles and lifts them straight up

30. Sitting beside a campfire on a still night, you feel warmth on the side of your face turned toward the fire, even though all the hot air streams straight up and away. If a thin

sheet of glass is held between your face and the flames, you still feel some warmth come through it. This shows the warmth reaches you mainly by:

- (A) Radiation, which travels in straight lines, needs no moving air, and can pass through transparent glass to reach your skin
- (B) Convection, because hot air must be curling around the glass to reach your face
- (C) Conduction, because the glass touches the air which touches your face in a chain
- (D) The fire pushing the cold night air toward you through the glass

31. Two identical black metal plates are at 400 °C and 100 °C respectively, both sitting in the same cool dark room. Comparing the heat each gives off as radiation every second:

- (A) The 100 °C plate radiates more, because cooler objects shed their heat faster to catch up
- (B) The 400 °C plate radiates much more heat per second, because the hotter an object is, the more radiation it emits
- (C) Both radiate the same amount, because they are identical plates of the same colour and size
- (D) Neither radiates, because radiation only happens in sunlight

32. A space probe far from any planet, surrounded by the vacuum of space, still slowly cools toward the cold of space. With no air or matter around it, by which mode does it lose this heat, and why are the others impossible?

- (A) By convection, since hot gases stream off the probe into the surrounding space
- (B) By conduction, since the vacuum directly carries heat away from the surface
- (C) It loses heat by radiation, since radiation needs no medium; conduction needs touching particles and convection needs a flowing fluid, neither of which exists in a vacuum
- (D) It cannot lose any heat at all, because all three modes need matter and a vacuum has none

33. Two identical kettles of equally hot water are placed in the same cool room: kettle M has a dull, dark, matt outer surface; kettle N has a bright, shiny, polished surface. After fifteen minutes, which kettle's water is cooler, and why?

- (A) Kettle M (dull dark surface) is cooler, because dull dark surfaces are better emitters of radiation and lose heat faster
- (B) Kettle N (shiny surface) is cooler, because shiny surfaces radiate heat away fastest
- (C) Both are equally cool, because they hold the same water at the same start temperature
- (D) Neither cools by radiation indoors, because radiation needs direct sunlight

34. A shiny silver teapot keeps tea hot longer than an identical dull black teapot of the same size, yet on a cold day a black object basking in the Sun warms up faster than a shiny one. Which single principle explains BOTH observations?

- (A) Dull dark surfaces are both better emitters AND better absorbers of radiation, while shiny surfaces are poorer at both — so black loses heat faster but also gains heat faster
- (B) Shiny surfaces both absorb and emit radiation better, which is why they keep tea hot and warm in sunlight
- (C) Dark surfaces absorb well but cannot emit, while shiny surfaces emit well but cannot absorb
- (D) Colour affects only absorption, never emission, so the teapot result must be due to something else

35. A student writes three claims about a perfect vacuum: '(1) conduction can cross it, (2) convection can cross it, (3) radiation can cross it.' Exactly how many are correct, and which?

- (A) Exactly two, because conduction and radiation both cross a vacuum but convection cannot
- (B) All three, because heat always finds some way across any gap
- (C) Exactly one — only claim (3), since radiation needs no medium, while conduction needs touching particles and convection needs a flowing fluid, both absent in a vacuum
- (D) None, because every mode of heat transfer needs matter to travel through

36. A frying pan needs a fixed handle. From the materials silver, aluminium, copper, and dry wood, which is correct for the handle, and why are the others wrong?

- (A) Dry wood, because it is a poor conductor and stays cool to grip; silver, aluminium, and copper are all metals that conduct heat well and would get hot
- (B) Silver, because as the best conductor it spreads heat out and so stays coolest
- (C) Aluminium, because light metals barely conduct heat and stay cool
- (D) Copper, because metal handles are the most comfortable and durable

37. A vacuum flask (thermos) keeps soup hot using a near-vacuum gap between two walls AND silvery shiny inner surfaces. Match each feature to the mode of heat loss it chiefly fights:

- (A) The vacuum gap blocks conduction and convection (both need matter to cross), while the shiny silvered walls cut radiation (shiny surfaces are poor emitters)
- (B) The vacuum blocks radiation, while the shiny walls block conduction and convection
- (C) The vacuum blocks all three modes by itself, and the shiny walls only make it look nice
- (D) The shiny walls block all three modes, and the vacuum gap only adds strength

38. A foam cup, a thin metal cup, and a ceramic mug of the same shape are each filled with the same hot drink in the same room. Ranking how well they keep the drink hot, from best to worst, the most reasonable order is:

- (A) Thin metal best, foam worst, because metal seals heat in by surrounding the drink
- (B) All three keep it equally hot, since they hold the same drink in the same room

(C) Foam best, ceramic in the middle, thin metal worst — foam's trapped air is the poorest conductor and metal conducts heat away fastest

(D) Ceramic best, foam worst, because solid walls block heat better than air pockets

39. Why is air, which is all around us, often the best insulator in clothing and yet a poor insulator on a windy day?

(A) Air is always a perfect insulator whether still or moving, because it has so few particles

(B) Still trapped air conducts heat very poorly and so insulates well, but moving air carries heat away by convection, so wind makes air a poor insulator

(C) Moving air conducts better than still air, which is why wind cools you

(D) Air destroys heat when still but creates heat when moving

40. A shiny silver-coloured rescue blanket is wrapped around a cold person to keep them warm. Unlike a thick wool blanket which works by trapping air, the shiny blanket keeps the person warm chiefly by:

(A) Generating warmth from the silver coating once it touches the skin

(B) Trapping a very thick layer of air inside the thin silver sheet

(C) Conducting the cold surroundings away from the person quickly

(D) Reflecting the person's own radiated body heat back toward them instead of letting it escape

41. Two people on a sunny beach wear shirts identical except for colour — one white, one black. The same two shirts are then worn at night facing a warm bonfire. In daytime sun the black-shirt wearer feels hotter; at night near the fire the black-shirt wearer ALSO loses body heat faster if they turn away from the fire. Which single fact explains both?

(A) Black absorbs well in the day but reflects well at night, switching its behaviour with the time

(B) White absorbs more during the day and emits more at night, so the colours swap roles

(C) Dark surfaces both absorb radiation better (so they soak up the Sun) and emit radiation better (so they shed body heat faster), while white reflects and emits poorly

(D) Colour only affects daytime comfort; the night-time difference must come from the cloth's thickness

42. A solar water heater on a roof uses a dark metal plate behind glass to warm water flowing through pipes. Why is the absorber plate painted dull black rather than shiny white?

(A) A dull black surface reflects the most sunlight, sending heat into the pipes by reflection

(B) A shiny white surface would absorb more, but black is chosen because it is cheaper

- (C) Colour does not matter for the plate; the glass alone does all the heating
- (D) A dull black surface absorbs the most of the Sun's radiation, heating the plate and the water in the pipes most effectively

43. In a hot desert town, one house is whitewashed (painted white) and an identical neighbour's house is painted dark brown, with the same thick walls and same windows. By mid-afternoon, which house is cooler inside, and why?

- (A) The white house is cooler inside, because its light surface reflects most of the Sun's radiation, so the walls absorb less heat
- (B) The dark brown house is cooler inside, because dark colours keep buildings cool by absorbing heat away from the rooms
- (C) Both are equally cool inside, because the walls are the same thickness
- (D) The white house is hotter inside, because white absorbs the most sunlight

44. On a cold but bright winter day, a market vendor wants to feel warmer while standing in direct sunlight. Choosing between an otherwise identical white coat and black coat, which is better, and what is the trade-off at night?

- (A) The white coat is warmer in the sun, because pale colours absorb more sunlight
- (B) The black coat is warmer in the daytime sun because it absorbs more solar radiation; but at night, away from the Sun, the black coat would also emit body heat away faster
- (C) The black coat is warmer at every time, because black always makes more heat
- (D) Colour makes no difference in the sun, since both coats are the same thickness

45. Two sealed boxes, one lined dull black inside and one lined shiny white inside, each hold a thermometer and sit in identical bright sunlight through a glass lid. After an hour the black-lined box reads much higher. If both boxes are now moved into a dark cool room, which box's thermometer falls fastest, and why?

- (A) The shiny white box cools fastest, because shiny surfaces lose heat by radiation fastest
- (B) Both cool at the same rate, because they are the same size and shape
- (C) Neither cools, because once heated a sealed box keeps its heat forever
- (D) The black-lined box cools fastest, because a dull dark surface is also a better emitter and radiates its heat away faster than the shiny one

46. During a hot, sunny afternoon at the coast, a windsock at the beach points steadily inland (from the sea toward the land). Combining the daytime breeze pattern with its cause, this is a:

- (A) Land breeze: the sea heats faster than the land by day, pushing air toward the shore
- (B) Sea breeze caused by the Sun directly dragging air landward across the water
- (C) Land breeze caused by the land staying cooler than the sea all afternoon
- (D) Sea breeze: the land heats faster than the sea by day, so air rises over the warm land and cooler air flows in from the sea

47. Late at night, well after sunset, a sailor near the coast finds a steady wind blowing from the shore out to the open sea. Identify the breeze and the temperature situation that drives it:

- (A) A sea breeze: the sea has cooled faster than the land, so air flows toward the open sea
- (B) A land breeze: the land stays warmer than the sea at night, pushing air seaward
- (C) A sea breeze caused by the night air automatically reversing the daytime wind
- (D) A land breeze: the land has cooled faster than the sea, so the sea is now warmer; air rises over the warm sea and cooler air flows out from the land

48. A coastal observer logs the wind for one full day. It blows inland from about 10 a.m. to 6 p.m. and out to sea from about 10 p.m. to 6 a.m., with quiet, shifting winds around dawn and dusk. The calm, shifting periods near dawn and dusk happen because:

- (A) Near dawn and dusk the land and sea are at nearly the same temperature, so there is little difference to drive air in either direction
- (B) The wind always stops for exactly two hours twice a day by a fixed rule
- (C) The Sun is directly overhead at dawn and dusk, cancelling all breezes
- (D) The sea and land are at their greatest temperature difference at dawn and dusk

49. A geography student argues that a sea breeze and a land breeze are completely different phenomena driven by different forces. A teacher replies that they are 'the same convection process running in opposite directions.' Why is the teacher correct?

- (A) Both are convection: air rises over whichever surface is warmer and cooler air flows in from the cooler surface; by day the land is warmer, by night the sea is warmer, so the same process simply reverses
- (B) The sea breeze is convection but the land breeze is conduction through the cooler ground at night
- (C) The sea breeze is driven by the Sun and the land breeze by the Moon, so they are different forces
- (D) They are different because the sea breeze is radiation and the land breeze is convection

50. On a still day with no land-sea temperature difference at all (an unusual, perfectly even-temperature coast), would a sea breeze or land breeze blow? Why?

- (A) A sea breeze, because the sea is always cooler than the land regardless of temperature
- (B) Neither — with no temperature difference, no surface is warmer to make air rise, so there is nothing to drive a breeze in either direction
- (C) Both at once, because the coast always produces some breeze
- (D) A land breeze, because nights always make air flow out to sea no matter the temperatures

51. Sea and land breezes are convection on a large coastal scale. In this convection, the rising air over the warmer surface is replaced by air flowing in along the surface from the cooler region. This means that, at any moment, the breeze you feel near the ground always blows:

- (A) From the warmer surface toward the cooler one, following the rising warm air
- (B) From the cooler surface toward the warmer one (sea to land by day, land to sea by night), filling in beneath the rising warm air
- (C) Always from the sea toward the land, both day and night
- (D) Straight upward only, since warm air rises vertically

52. Wearing three thin shirts can feel warmer than one thick shirt of the same total cloth thickness. Each extra layer adds an extra thin gap. The main reason this layering is warmer is:

- (A) Three shirts weigh more than one, and heavier clothing is always warmer
- (B) Each gap between layers traps an extra pocket of still air, and trapped still air is a poor conductor, so more trapped-air pockets slow heat loss more
- (C) Three shirts produce three times as much body heat as one
- (D) Three thin layers conduct the body's heat outward faster, which feels warm

53. A down sleeping bag squashed flat under a heavy weight keeps a camper far less warm than the same bag fluffed up, even though it contains exactly the same amount of down. Why?

- (A) Fluffing the down traps thick pockets of still air, the real insulator; squashing it flat removes the trapped air, so heat escapes faster
- (B) Squashing the down stops it producing heat, so the bag goes cold
- (C) The squashed down conducts the body's heat away faster because it is now thicker
- (D) A flat bag reflects less body heat, so the camper loses heat by reflection

54. A camper has a roll of shiny aluminium foil and a thick woollen blanket, and wants to keep a hot pot of stew warm as long as possible. The best approach is:

- (A) Wrap the thick woollen blanket directly around the pot, because wool traps still air and is a poor conductor that strongly slows heat loss
- (B) Wrap the aluminium foil directly around the pot, because metal foil traps heat by sealing it inside
- (C) Use only the foil and leave the wool aside, because thin metal insulates better than thick wool
- (D) Either choice works equally well, since both feel warm when held

55. Order these four cups of identical hot tea, in the same room, from the one that stays hot LONGEST to the one that cools soonest: (i) a foam cup, (ii) a thin uncovered metal cup, (iii) a foam cup with a lid, (iv) a thin metal cup wrapped in wool. The correct order is:

- (A) Uncovered thin metal cup longest, then plain foam cup, then wool-wrapped metal, then foam-with-lid soonest
- (B) Foam cup with a lid longest, then wool-wrapped metal cup, then plain foam cup, then uncovered thin metal cup soonest
- (C) All four cool at the same rate, since they hold the same tea in the same room
- (D) Plain foam cup longest, then foam-with-lid, then uncovered metal, then wool-wrapped metal soonest

56. A student lists how heat reaches them in three situations and the mode involved: 'Toes warmed by a hot-water bottle touching them — conduction. Warmth felt across a room from a glowing heater I am not touching — radiation. Warm air spreading from a heater placed near the floor to fill a room — convection.' How many of these three pairings are correct?

- (A) All three are correct
- (B) Only two are correct
- (C) Only one is correct
- (D) None is correct

57. A thermos of hot cocoa is left in a cold room for hours and eventually goes lukewarm despite its vacuum walls and shiny lining. Which statement best explains why a thermos only SLOWS heat loss rather than stopping it forever?

- (A) A thermos generates a little heat of its own, which runs out after a few hours
- (B) A vacuum actually speeds up heat loss, so the cocoa cools as fast as in any cup
- (C) Cold from the room flows into the cocoa through the vacuum, which a thermos cannot block
- (D) The vacuum and shiny walls greatly slow conduction, convection, and radiation, but the cap, seams, and tiny radiation through the walls still let heat leak slowly, so it cannot be perfect

58. Consider this chain on a sunny coastal afternoon: (1) the dark land absorbs sunlight by radiation and warms faster than the sea; (2) the air just above the warm land is heated and rises; (3) cooler sea air flows in to replace it. Identify the heat-transfer mode in step 1 and in steps 2–3, in that order:

- (A) Step 1 is conduction (the Sun conducts heat to the land); steps 2–3 are radiation (the warm air beams to the sea)
- (B) Step 1 is convection (hot solar gas reaches the land); steps 2–3 are conduction (the air conducts heat sideways)
- (C) Step 1 is radiation (Sun's heat crossing space and absorbed by the land); steps 2–3 are convection (warm air rising and cooler air flowing in)
- (D) All three steps are conduction, since the ground, air, and sea are all in contact

59. A laboratory thermometer reads 25 °C in a room. A student grips its bulb tightly in a warm hand for a minute, the thread rises to about 33 °C, then they let go and watch. What does the thread do, and what does this show about how the thermometer works?

- (A) The thread stays at 33 °C, because a thermometer always keeps its highest reading
- (B) The thread keeps rising past 33 °C, because the hand keeps adding heat after letting go
- (C) The thread drops to 0 °C, because a thermometer with no hand on it reads zero
- (D) The thread falls back toward 25 °C, showing the liquid expands when warmed by the hand and contracts again as it cools, with no kink to hold it

60. On a winter night a thin metal sheet and a thick foam board, left outdoors together, both reach the air temperature of 2 °C. A barefoot person steps briefly on each. Which feels colder underfoot, and would a thermometer pressed on each read differently?

- (A) The foam feels colder, and a thermometer would read a lower temperature on the foam
- (B) The metal feels much colder underfoot because it conducts heat from the foot quickly, yet a thermometer would read 2 °C on both since they are at the same temperature
- (C) Both feel equally cold, and a thermometer reads 2 °C on both, since feel and temperature always match
- (D) The metal feels colder, and a thermometer would read a lower temperature on the metal, confirming it is genuinely colder

Solutions — Science (Class 7), Chapter 4: Heat — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	B	31	B	41	C	51	B
2	D	12	B	22	D	32	C	42	D	52	B
3	A	13	B	23	A	33	A	43	A	53	A
4	B	14	B	24	D	34	A	44	B	54	A
5	C	15	D	25	D	35	C	45	D	55	B
6	C	16	C	26	D	36	A	46	D	56	A
7	C	17	C	27	C	37	A	47	D	57	D
8	A	18	D	28	A	38	C	48	A	58	C
9	B	19	C	29	B	39	B	49	A	59	D
10	C	20	C	30	A	40	D	50	B	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Heat flows from the hotter water to the cooler water until both reach one common temperature. With equal amounts of the same liquid, that common value sits exactly halfway between the two starting temperatures: $(20 + 80) \div 2 = 50$ °C. The hotter portion does not simply impose its 80 °C — it cools as it warms the other half. Adding the two temperatures (giving 100) treats temperature like a quantity that piles up, which it does not. And heat does not settle at the lower value of 20 °C; the cool water warms up while the hot water cools down.

2. (D) When the amounts are unequal, the final temperature is pulled toward the temperature of the larger amount. Here there is three times as much hot water (300 g) as cold water (100 g), so the mix ends up well above the midpoint of 50 °C and much closer to 80 °C. The midpoint of 50 °C would only be right if the amounts were equal. 'Cold water always wins' is false — the side with more substance has the bigger say, and here that is the hot water. Plainly averaging the two temperatures and nudging the result also ignores that the larger hot share controls the outcome.

3. (A) The final temperature is dragged toward whichever side has far more substance. The bucket has overwhelmingly more water than one drop, so the bucket barely changes and stays near 25 °C; the drop's small heat is shared out among a huge mass of water. The midpoint of 62.5 °C would only apply if the two amounts were equal, which they are wildly not. Being the hottest object does not let one drop heat a whole bucket — heat content depends on amount as well as temperature. And adding warm water cannot cool the bucket below 25 °C.

4. (B) Heat flows only when there is a temperature difference, always from hotter to colder. Both blocks are at 40 °C, so there is no difference and no net flow — both simply stay at 40 °C. Having more metal gives block X more total heat, but the same temperature, so it does not push heat into Y. 'More metal means higher temperature' confuses amount of heat with temperature; they are different. And although X stores more heat than Y, heat does not flow without a temperature gap.

5. (C) Heat travels from the hotter body to the cooler one. The tea is hotter than the room, so heat moves out of the tea into the room air; the tea cools and the room warms by a tiny, barely noticeable amount, until the tea matches room temperature. The student's claim reverses this — the room does not warm the tea. There is no separate substance called 'cold' that flows; cooling is just the loss of heat. And size does not set the flow direction — temperature difference does.

6. (C) For the same liquid and the same amount, the heat needed depends on how many degrees the temperature rises, not on the starting value. Both rose by 10 °C, so each absorbed about the same heat. Ending at a higher temperature (Q at 80 °C) does not by itself mean more heat was added during this step. Starting cold does not make P need extra heat for the same 10 °C climb. And heat is exactly what raised each temperature, so saying no heat entered is wrong — heat and temperature change are tightly linked here.

7. (C) Task (i) is at -5 °C, far below the clinical scale's lowest mark of 35 °C, so only the wide-range laboratory thermometer can reach it. Task (ii) needs a reading held in place after the thermometer leaves the body and carried to another room — that is exactly what the clinical thermometer's kink does, plus its scale is spread finely over the body range for tenth-of-a-degree fevers. Using the clinical one for both fails on (i) because -5 °C is off its scale. Using the laboratory one for both fails on (ii) because its kink-free thread would slip as it cooled on the walk to the other room. The last pairing assigns the wrong tool to each task.

8. (A) The kink traps the thread at the highest temperature it has reached. After the first patient it sits at 39.4 °C. The second patient at 38.6 °C is cooler than 39.4 °C, so the liquid does not expand enough to push the trapped thread any higher — it stays near 39.4 °C, giving a falsely high reading. This is precisely why a clinical thermometer must be shaken down before each use. The thermometer does not reset itself; that is the job of the shake.

Readings do not add up across uses. And the kink stops the thread from dropping back, so it does not fall to room temperature on its own.

9. (B) A laboratory thermometer must cover about 120 degrees of range, so each small division is comparatively large (around 1 °C), and it cannot distinguish 39.2 °C from 39.3 °C. A clinical thermometer devotes its whole length to roughly 35–42 °C, so it shows tenths of a degree clearly. The range claim is false: 39.2 °C is inside –10 to 110. Lacking a kink only means the reading is not held while carried away — it can still measure body temperature in contact. And the liquid does not freeze anywhere near body temperature.

10. (C) A laboratory thermometer has no kink, so its thread moves freely up and down with temperature; once removed into cooler air the liquid contracts and the reading falls. A clinical thermometer's kink traps the thread, so its reading stays put after removal. It is not true that every thermometer locks its reading — only the kink does that. The kink-free laboratory thermometer is precisely the one that does NOT hold its reading. And neither reading climbs after removal, since the surroundings are cooler than 100 °C.

11. (D) Across 100 °C, A's 100 divisions make each one $100 \div 100 = 1$ °C; B's 50 divisions make each one $100 \div 50 = 2$ °C. The thermometer whose smallest division is smaller can be read more finely, so A (1 °C steps) beats B (2 °C steps). Fewer divisions over the same range make each division LARGER, not smaller, so B is coarser. Equal ranges do not guarantee equal fineness — division count decides it. And the computed values are 1 °C and 2 °C, not 0.5 °C and 1 °C.

12. (B) Ten equal divisions in one degree make each division $1 \div 10 = 0.1$ °C. Seven divisions above 38 means $38 + 7 \times 0.1 = 38.7$ °C. Reading 38.07 places the decimal one digit too far, as if each division were 0.01 °C. Counting the 7 divisions as whole degrees ($38 + 7 = 45$) ignores that they are tenths. And 38.35 would require five divisions per degree of 0.05 each, which is not the stated scale.

13. (B) Twenty degrees split into 10 equal divisions makes each division $20 \div 10 = 2$ °C. Six divisions above 40 means $40 + 6 \times 2 = 40 + 12 = 52$ °C. Reading 46 wrongly treats each division as 1 °C ($40 + 6$). Reading 40.6 wrongly treats each division as 0.1 °C. And 60 jumps to the next numbered mark, ignoring that the thread is only 6 of the 10 divisions up.

14. (B) Five degrees over 5 divisions makes each division 1 °C. Yesterday: $10 + 4 = 14$ °C. Today: $15 + 2 = 17$ °C. The change is $17 - 14 = +3$ °C, a rise. Reading today's value as 16 forgets the reading starts from the 15 mark plus 2 = 17, not $14 + 2$. Comparing division counts (4 vs 2) and concluding a fall ignores that today's count is measured above a higher numbered mark (15, not 10), so the temperature actually rose. And 5 °C is just the gap between the numbered marks, not the change in the thread's position.

15. (D) Ten divisions per 10 degrees make each division 1 °C. Seven divisions below 0, moving toward colder, gives $0 - 7 = -7$ °C. Reading +7 °C ignores that the thread is below

zero, in the colder direction. Reading -70 wrongly treats each division as 10°C . And -17 wrongly starts counting from the -10 mark instead of from 0.

16. (C) Two degrees split into 4 divisions makes each division $2 \div 4 = 0.5^{\circ}\text{C}$. Three divisions above 24 means $24 + 3 \times 0.5 = 24 + 1.5 = 25.5^{\circ}\text{C}$, so the second student is correct. The first student assumed each division was 0.1°C , which is only true when there are 10 divisions per degree — not the case here. The two values 24.3 and 25.5 are clearly different numbers. And 27°C would require each division to be 1°C with three of them, which contradicts the stated 0.5°C divisions.

17. (C) Each division is 1°C , so the readings are $30 + 2 = 32$, $30 + 5 = 35$, $30 + 8 = 38^{\circ}\text{C}$ — a steady rise of 3°C per minute. Continuing the pattern, 9:03 gives $38 + 3 = 41^{\circ}\text{C}$, which is 1 division above the 40 mark. The thread is not capped at the 9th division above 30; it simply crosses into the next numbered region. The rise is steady (adding 3 each minute), not doubling. And there is no rule forcing the thread to stop on a numbered mark — it lands wherever the temperature is.

18. (D) In a solid, heat travels by conduction: vibrating particles pass energy to their neighbours, so the disturbance reaches near points first and far points later. That is exactly why the wax falls in sequence from the flame end outward — a gradual arrival, not chance. If radiation reached every bead equally and instantly, they would fall together, which they do not. Convection needs a fluid that can flow; the solid rod's particles are locked in place and cannot circulate. And the wax does not make its own heat — it merely melts when conducted heat arrives.

19. (C) The faster the far bead melts, the faster heat reached the end, so the best conductor loses its bead first. Copper (30 s) is fastest, glass (5 min) is in the middle, wood (15 min) is slowest — so the order is copper > glass > wood. Being light does not improve conduction; wood is light yet a poor conductor. Glass conducts far less than copper, so it is not best. And an identical heat source does NOT equalise the results — the material decides how quickly heat passes through, which is the whole point of the experiment.

20. (C) Same shape means the only difference is the material. Copper is a good conductor and quickly delivers heat to its far end; wood is a poor conductor, so heat creeps along it slowly and the far end needs much more time to reach the same temperature. Identical shape does not make the times equal, because the materials conduct at very different rates. Poor conductors heat their far ends MORE slowly, not faster. And wood does conduct heat — just slowly — so it is not 'never.'

21. (B) Both objects are at 10°C , so neither is actually colder. But the metal knob is a good conductor: it whisks heat away from your warmer hand quickly, cooling the skin fast, so it FEELS colder. Wood is a poor conductor and draws heat slowly, so it feels less cold. The wood is not at a lower temperature — both are 10°C . They do not feel equally cold despite

equal temperature, because conduction rates differ. And the metal, being colder than your hand, takes heat from it rather than giving heat back, so it cannot feel warmer.

22. (D) Conduction is the passing of energy from particle to neighbouring particle. In copper the particles are tightly packed and constantly in contact, so energy is relayed quickly. In air the particles are widely separated and seldom touch, so the relay is feeble and slow. Air certainly contains particles — they are just spread out. Temperature does not set conduction ability; copper is not 'colder.' And air particles actually move fast; it is their large spacing, not slowness, that makes conduction poor.

23. (A) The body must carry heat from the burner into the food, which calls for a good conductor — metal. The handle must keep heat from reaching the hand, which calls for a poor conductor (insulator) — wood or plastic. Reversing them (wooden body, metal handle) would leave the food barely warmed while the handle conducts heat and burns the hand. An all-metal pan would make the handle hot, not cool. And an all-plastic pan would not heat the food well, since plastic is a poor conductor — defeating the cooking goal.

24. (D) Heated water expands, so the same mass takes more space, making it less dense (lighter) — it rises. Cooler, denser water sinks to replace it, and the cycle repeats, forming a convection current that warms the whole kettle. Hot water is lighter, not heavier, so it does not sink. Water conducts heat poorly, so the heat is not simply conducted straight up through still layers — the bulk movement of water does most of the carrying. And the water does not flash to steam at ordinary heating; it circulates as a liquid first.

25. (D) Convection currents rise wherever the water is heated, because that is where the water expands, lightens, and floats upward; the cooler water elsewhere sinks. Heating a corner therefore anchors the rising limb at that corner, not the centre. The current does not always rise centrally — it follows the heat source. Conduction through still water is far too slow to trace this loop; the dye moves because the water bulk-moves. And heated water becomes lighter, so it rises at the corner, not sinks.

26. (D) Warm air rises. A floor-level heater sends warm air up and around the whole room as a convection loop. A ceiling-level heater would warm air that is already at the top, where it tends to stay, leaving the lower (living) part of the room cold — so low placement heats a room better. Warm air does not fall down to fill the room; that reverses convection. Air conducts heat poorly, so heating does not spread by conduction regardless of height. And warm air rises, so the floor would not become the hottest part.

27. (C) Convection works only in fluids (liquids and gases) because the warmed parts must rise and circulate, carrying heat bodily. In a solid the particles are locked in their positions and cannot flow, so no convection current can form — heat spreads through iron by conduction instead. Paint colour affects radiation, not whether a solid can convect. No amount of heat lets a rigid solid's particles circulate. And solids certainly can be heated in the middle; they just transfer that heat by conduction.

28. (A) Convection is driven by warm (lighter) fluid rising and cold (denser) fluid sinking. Here the warm water is already on top and the cold below, which is already the stable layering, so there is no tendency to overturn and circulate. Warm water is lighter, not denser, so it belongs on top — the arrangement is stable, not blocked. Convection can certainly occur in tall cylinders when heated from below; shape is not the issue. And the cold layer does not 'block' cooling — there is simply no driving force for a current.

29. (B) The dark bare ground absorbs more sunlight and becomes hotter than the shaded grass; it warms the air just above it more strongly, and that hotter air expands, becomes lighter, and rises as a vigorous convection current, carrying the smoke up. Over the cooler shaded patch the air rises only weakly. Smoke's lift here comes from the rising warmed air, not from smoke being inherently lighter over one surface. Conduction does not bodily push air upward. And radiation travels in straight rays and does not lift particles by buoyancy — the buoyant rising air does that.

30. (A) The hot air rises away from you, so convection cannot be warming your face. The warmth that crosses the gap — and even passes through a sheet of glass — is radiation: it travels in straight lines, needs no medium, and transparent glass lets much of it through. Convection is ruled out because the hot air goes up, not around the glass. Air is a poor conductor and the gap is large, so conduction is negligible — and a chain of touching air does not deliver this warmth. And the fire does not warm you by shoving air through glass; the radiation does.

31. (B) Every warm object emits radiation, and the hotter it is, the more it emits each second. So the 400 °C plate radiates far more than the 100 °C plate. A cooler object does not shed heat faster — it emits less. Identical size and colour do not equalise emission when the temperatures differ so much; temperature is the dominant factor here. And objects radiate their own heat regardless of sunlight, so they radiate even in a dark room.

32. (C) In a vacuum there are no particles to touch (so no conduction) and no fluid to flow and circulate (so no convection). Radiation, however, needs no medium and travels through empty space, so the probe sheds heat by radiating it away. There is no surrounding gas to carry convection. There is no matter to conduct through. And it is wrong that no heat can leave — radiation works precisely where the other two fail, which is why bodies in space still cool.

33. (A) A dull, dark surface is a good emitter of radiation, so kettle M radiates its heat away faster and its water ends cooler. A shiny, polished surface is a poor emitter, so kettle N holds its heat longer — it cools slower, not faster. Equal starting water does not make their cooling equal, because the surface type changes how fast each radiates. And objects radiate their own heat regardless of sunlight, so they lose heat by radiation indoors too.

34. (A) A dull dark surface is good at both emitting and absorbing radiation; a shiny surface is poor at both. So the black teapot loses (emits) heat faster, keeping tea hot for

less time, while in sunlight a black object gains (absorbs) heat faster and warms quicker. One property — dark surfaces are strong emitters and absorbers — covers both facts. Shiny surfaces are poor, not good, at both, so they cannot explain it the reverse way. Dark surfaces both absorb and emit well, not one without the other. And colour affects emission too, not absorption alone, so the teapot result is fully explained by surface type.

35. (C) A vacuum has no particles in contact (so conduction fails) and no fluid to flow (so convection fails). Radiation, needing no medium, crosses freely — so exactly one claim, number (3), is correct. Conduction does NOT cross a vacuum, so 'two' is wrong. Not all three work — only radiation does. And 'none' wrongly denies radiation, which genuinely does cross empty space (that is how the Sun warms the Earth).

36. (A) A good handle must NOT pass heat to the hand, so it should be a poor conductor (insulator) — dry wood fits. Silver, aluminium, and copper are all metals and good conductors, so any of them would carry the pan's heat to the grip and burn the hand. Being the best conductor (silver) makes the handle hottest, not coolest. Aluminium, though light, is still a metal and conducts heat well — 'light metals barely conduct' is false. And comfort or durability does not override the fact that a copper handle would get dangerously hot.

37. (A) Conduction needs touching particles and convection needs a flowing fluid, so removing the matter between the walls (a vacuum) chokes off both of those paths. Radiation, however, crosses a vacuum, so it must be fought separately — by making the surfaces shiny, since shiny surfaces are poor emitters of radiation. Reversing the two (vacuum stops radiation, shiny stops conduction) is backwards. The vacuum alone cannot stop radiation, so it does not block all three. And the shiny walls do not block conduction or convection, so they do not handle all three either.

38. (C) Insulation depends on how poorly the wall conducts. Foam is full of trapped air pockets, the poorest conductor, so it keeps the drink hot longest. A thin metal cup conducts the drink's heat outward fastest, so it cools the drink soonest. Ceramic sits between them. Metal does not 'seal heat in' — it conducts heat out quickly, the opposite. Same drink and room do not equalise them, since the wall materials conduct very differently. And trapped air in foam is a BETTER insulator than a solid ceramic wall, so foam beats ceramic.

39. (B) Air's widely spaced particles conduct heat very poorly, so when air is trapped and held still (as in wool or a sweater) it barely lets heat through — an excellent insulator. But when air moves (wind), it carries heat away bodily by convection, so it no longer insulates and instead cools you. So air is not always a perfect insulator — only when still. Moving air does not conduct better; it removes heat by convection, a different mechanism. And air neither destroys nor creates heat; it only transmits or carries it.

40. (D) A thin shiny sheet has almost no trapped air, so it cannot insulate the way wool does. Instead, its shiny surface is a poor emitter and a good reflector of radiation, so it

reflects the body's own radiated heat back toward the person, slowing radiative heat loss. The blanket does not generate warmth — no material makes heat from being touched. A thin sheet does not trap a thick air layer, so that is not its mechanism. And 'conducting the cold away' is backwards: there is no flow of cold, and conducting heat outward would cool the person.

41. (C) A dark surface is a strong absorber AND a strong emitter of radiation. By day it absorbs more sunlight, so the black shirt feels hotter; at night, facing away from the fire, that same strong-emitting black surface radiates the wearer's body heat away faster. One property explains both. A surface does not switch from absorbing to reflecting by time of day — its absorbing/emitting nature is fixed. White does not absorb more by day; it reflects more. And the night effect is from emission, not cloth thickness, since the shirts are identical except for colour.

42. (D) The job of the absorber plate is to take in as much solar radiation as possible and pass that heat to the water. Dull dark surfaces are the best absorbers of radiation, so black absorbs the most sunlight and heats the water most effectively. Black absorbs rather than reflects, so 'reflects the most' is wrong. White reflects more and absorbs less, so it would be worse, not better — and the choice is about absorption, not cost. And the colour of the plate matters greatly; the glass merely lets sunlight in and traps warmth, it does not replace the absorbing surface.

43. (A) A white surface reflects most of the Sun's radiation, so the white house's walls absorb less heat and the rooms stay cooler. A dark surface absorbs much more, so the brown house's walls take in more heat and the rooms get hotter. 'Dark colours keep buildings cool by absorbing heat' is self-contradictory — absorbing heat warms the building. Equal wall thickness does not equalise them, because the surface colour changes how much radiation is absorbed. And white reflects, not absorbs, so the white house is cooler, not hotter.

44. (B) In direct sunlight a dark coat absorbs more of the Sun's radiation, so the black coat warms the wearer more by day. The catch is that a dark surface is also a better emitter, so at night (no Sun) it would shed the body's own radiated heat faster — a genuine trade-off. White reflects sunlight, so it absorbs less and feels cooler, not warmer, in the sun. Black does not 'make heat'; it absorbs the Sun's radiation, so it is not warmer at every time. And colour does affect absorption regardless of equal thickness.

45. (D) A dull dark surface is a strong absorber (so it got hotter in the sun) AND a strong emitter (so it now radiates its heat away faster in the dark). The black-lined box therefore both heated up more and cools down faster. Shiny surfaces are poor emitters, so the white-lined box loses heat slowly, not fastest. Equal size and shape do not equalise cooling, because the surface type sets the emission rate. And no sealed box keeps heat forever — it always radiates heat away to cooler surroundings.

46. (D) By day the land heats faster than the sea. The air over the warm land rises, and cooler air flows in from over the sea to replace it — that inflow from sea to land is the sea breeze. It is not a land breeze, and the sea does not heat faster than land by day. The Sun does not physically drag air sideways; uneven heating drives the breeze. And the land is the WARMER surface in daytime, not cooler, so the cause is land heating faster, not staying cool.

47. (D) At night the land loses heat and cools faster than the sea, so the sea ends up warmer. Air rises over the warmer sea and cooler air flows out from the cooled land toward the sea — a land breeze (land to sea). It is not a sea breeze, and the sea does not cool faster than land. The land is COOLER than the sea at night, not warmer, so 'land stays warmer' is wrong. And the reversal is caused by the land-sea temperature swap, not by an automatic rule.

48. (A) The breeze is driven by the temperature difference between land and sea. By day the land is warmer (sea breeze); by night the land is cooler (land breeze). Near dawn and dusk the land and sea are passing through nearly equal temperatures, so the difference shrinks to almost nothing and the driving force fades — winds go quiet and shift. There is no fixed two-hour-stop rule; it depends on the temperatures matching. The Sun is low near the horizon at dawn and dusk, not overhead. And the temperature difference is smallest, not greatest, at those changeover times — that is exactly why the wind is weak then.

49. (A) Both breezes are the same convection mechanism: air rises over the warmer surface and cooler air flows in from the cooler surface. The only thing that changes between day and night is which surface is warmer — the land by day, the sea by night — so the wind simply reverses direction. The same process runs both ways, confirming the teacher. The land breeze is not conduction; air conducts heat poorly, and the breeze is bulk air movement. The Moon does not drive the land breeze; uneven heating and cooling do. And neither breeze is radiation — both are convection of air.

50. (B) Sea and land breezes are convection driven by a temperature difference between land and sea: air rises over the warmer surface and cooler air flows in. If both are at exactly the same temperature, no surface is warmer, no air preferentially rises, and so neither breeze can form. The claim that the sea is 'always cooler' is false — by night the sea is the warmer one. A coast does not always have a breeze; without a temperature difference there is none. And nights produce a land breeze only because the land cools below the sea — with no difference, even that does not occur.

51. (B) Warm air rises over the warmer surface, leaving a gap near the ground that is filled by air flowing in from the cooler surface. So the surface breeze always moves FROM the cooler region TOWARD the warmer one: sea-to-land by day (land warmer) and land-to-sea by night (sea warmer). It does not blow from warm to cool at ground level — that air has risen, not flowed sideways. It is not always sea-to-land; the direction reverses at night.

And while warm air does rise vertically, the breeze we feel is the horizontal inflow replacing it, not a purely upward wind.

52. (B) The warmth of clothing comes mainly from trapped still air, which conducts heat poorly. Each gap between layers traps another pocket of still air, so three layers create more insulating air pockets than one thick layer of the same cloth, slowing heat loss more. Weight does not set warmth — trapped air does. Clothes do not generate body heat, so three shirts do not 'produce' more heat. And conducting heat outward faster would cool you, the opposite of feeling warm.

53. (A) Down insulates by holding many pockets of still air, which conduct heat very poorly. Fluffing it up maximises those air pockets; squashing it flat squeezes the air out, so there is little trapped air left and body heat escapes quickly. The down does not produce heat, so 'stops producing heat' is wrong. A squashed bag is thinner, not thicker, and the loss is from losing trapped air, not from the down conducting better. And the warmth is from trapped air slowing conduction, not from reflecting body heat.

54. (A) Wool is full of trapped still air and is a poor conductor, so wrapping it around the pot greatly slows the stew's heat from escaping — the best insulation here. Aluminium foil is a metal and a good conductor, so it would carry heat from the pot to its outer surface quickly; it does not 'seal heat in' by blocking it. Thin metal does not insulate better than thick wool — it conducts heat away faster. And the two are not equally effective, since wool insulates far better than metal foil.

55. (B) Heat is kept by poor conductors (trapped air) and by covering openings to stop convection/evaporation from the top. The foam cup with a lid combines foam's trapped air with a lid that blocks heat escaping from the open surface, so it stays hottest. The wool-wrapped metal cup gains good insulation from the wool's trapped air despite the conducting metal, so it is next. A plain foam cup insulates the sides but loses heat from the open top. An uncovered thin metal cup conducts heat out fastest and is open on top, so it cools soonest. The cups do not cool equally, because their walls and coverings differ. And the lid plus foam beats plain foam, so 'plain foam longest' is wrong.

56. (A) Touching a hot-water bottle transfers heat by direct particle contact — conduction (correct). Feeling warmth across a room from a glowing heater without touching it, through air that may not even be moving toward you, is radiation travelling in straight lines (correct). Warm air rising from a floor heater and circulating to fill the room is bulk movement of a heated fluid — convection (correct). All three pairings match their definitions, so the count is three. Each of the smaller-count options would require at least one pairing to be wrong, but none of them is.

57. (D) A thermos drastically reduces all three modes — the vacuum starves conduction and convection of the matter they need, and the shiny walls cut radiation — but it cannot remove every path. Heat still leaks slowly through the cap, the seams where the walls join,

and by some radiation, so over hours the cocoa drifts to room temperature. The thermos does not generate heat of its own to 'run out.' A vacuum slows heat loss, it does not speed it up. And there is no flow of 'cold' into the cocoa — cooling is the slow escape of the cocoa's own heat to the cooler room.

58. (C) Step 1: the Sun's heat crosses the vacuum of space and is absorbed by the dark land — that crossing of empty space and absorption is radiation. Steps 2–3: the heated air over the land rises and cooler sea air flows in to replace it — bulk movement of a fluid, which is convection. The Sun cannot conduct heat through space (no matter to conduct through), so step 1 is not conduction. Warm air rising and flowing is not radiation, so steps 2–3 are not radiation. And the moving air is convection, not conduction, since air conducts heat poorly and here it is physically flowing — so 'all conduction' is wrong.

59. (D) A laboratory thermometer has no kink, so its thread moves freely. The warm hand heats the bulb's liquid, which expands and pushes the thread up to about 33 °C; once the hand is removed the liquid cools toward room temperature, contracts, and the thread falls back toward 25 °C. This demonstrates the expand-on-heating, contract-on-cooling behaviour the thermometer relies on. Only a clinical thermometer's kink holds the highest reading — this one does not. The thread cannot keep rising once the hand is gone, since no more heat is being added. And with no hand on it the thermometer reads the room's 25 °C, not 0 °C.

60. (B) Both objects are at 2 °C, so a thermometer reads 2 °C on each — they are equally cold by measurement. But the metal is a good conductor and pulls heat from the warm foot rapidly, chilling the skin fast, so it FEELS far colder; the foam, a poor conductor, draws heat slowly and feels less cold. Foam does not feel colder, and it is not at a lower temperature. Feel and temperature do not always match — that mismatch is the whole point, since the metal feels colder despite equal temperature. And the thermometer would NOT read lower on the metal, because the metal is not genuinely colder — it only conducts heat away faster.